Ficha 1

1. $\overrightarrow{P_1P_2} = ?$

a) $P_1 = (1, 2, 1)$ $P_2 = (-5, 3, 1)$

$P_2 - P_1 = \overrightarrow{P_1P_2} = (-6, 1, 0)$

b) $P_1 = (-3, 2, -1)$ $P_2 = (15, 2, 6)$

$P_2 - P_1 = \overrightarrow{P_1P_2} = (18, 0, 7)$

c) $P_1 = (12, 22, 1)$ $P_2 = (5, 23, 11)$

$P_2 - P_1 = \overrightarrow{P_1P_2} = (-7, 1, 10)$

2. $\|\vec{u}\| = \sqrt{(1)^2 + (2)^2 + (3)^2} = \sqrt{1 + 4 + 9} = \sqrt{14}$

$\|\vec{v}\| = \sqrt{(-1)^2 + (2)^2} = \sqrt{1 + 4} = \sqrt{5}$

$\|\vec{w}\| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$

3. $\vec{u} \cdot \vec{v} = 0$ são ortogonais

a) $(1, 2, -1) \cdot (-5, 3, 1) = -5 + 6 - 1 = 0$ Condição verificada

b) $(-3, 2, 1) \cdot (1, 2, -6) = -3 + 4 - 6 = -5$ condição não verificada

c) $(2, -2, 2) \cdot (-2, 2, 1) = -4 - 4 + 2 = -6$ Condição não verificada

4. $(1, 2, 3) \cdot (2, k, 4) = 0 \Leftrightarrow (2 + 2k + 12) = 0 \Leftrightarrow$

$\Leftrightarrow k = -7$

5. $\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & -1 \\ -5 & 3 & 1 \end{vmatrix} = \left(\vec{i} \begin{vmatrix} 2 & -1 \\ 3 & 1 \end{vmatrix}, \vec{j} \begin{vmatrix} 1 & -1 \\ -5 & 1 \end{vmatrix}, \vec{k} \begin{vmatrix} 1 & 2 \\ -5 & 3 \end{vmatrix} \right) =$

$= (5, +4, 13)$

6. $\|\vec{u}\times\vec{v}\| = \sqrt{(5)^2+(-4)^2+(13)^2} = \sqrt{25+16+169} =$
$= \sqrt{210}$

7.
$\vec{u} = \vec{v} + \gamma\vec{w} \Leftrightarrow$ $\qquad$ $\cancel{\vec{v}}\times\vec{w} = \cancel{\vec{v}}\times\vec{w} \Leftrightarrow$
$\Leftrightarrow \vec{u} = \vec{v} - (\vec{u} - \vec{v})$ $\qquad$ $\Leftrightarrow \cancel{\vec{w}} = \cancel{\vec{w}} \Leftrightarrow$
$\Leftrightarrow 2\vec{u} = 2\vec{v}$ $\qquad$ $\Leftrightarrow 0 = 0$ c.q.m

9/8. $$\left|\begin{vmatrix} 1 & 2 & -1 \\ -5 & 3 & 1 \\ 1 & 1 & 0 \end{vmatrix}\right| = \left| -\begin{vmatrix} 1 & -1 \\ -5 & 1 \end{vmatrix} + \begin{vmatrix} 2 & -1 \\ 3 & 1 \end{vmatrix} \right| =$$

$$= |-1+5+2+3| = |9| = 9$$

$\boxed{\text{ou}}$

$$\begin{vmatrix} i & j & k \\ 1 & 2 & -1 \\ -5 & 3 & 1 \end{vmatrix} = (5, -4, 13)$$

$$|(\vec{u}\times\vec{v})\cdot\vec{w}| = |(5,-4,13)\cdot(1,1,0)| = 9$$

10. $$\begin{vmatrix} 1 & 2 & -3 \\ 1 & K & 1 \\ 3 & 2 & 1 \end{vmatrix} = \begin{vmatrix} K & 1 \\ 2 & 1 \end{vmatrix} - \begin{vmatrix} 2 & -3 \\ 2 & 1 \end{vmatrix} + 3\begin{vmatrix} 2 & -3 \\ K & 1 \end{vmatrix} = 0 \Leftrightarrow$$

$$= K - 2 - 2 - \cancel{6+6} + 9K = 0 \Leftrightarrow$$
$$\Leftrightarrow 10K = 4 \Leftrightarrow$$
$$\Leftrightarrow K = \frac{4}{10}$$

11. $\overrightarrow{P_1P} = P - P_1 = (x-5, y, z-7)$

$\overrightarrow{PP_2} = P_2 - P = (2-x, -3-y, 6-z)$

$\overrightarrow{P_1P} = 3\overrightarrow{PP_2} \Leftrightarrow (x-5, y, z-7) = (6-3x, -9-3y, 18-3z) \Leftrightarrow$
$(x, y, z) = \left(\frac{11}{4}, -\frac{9}{4}, \frac{25}{4}\right)$

12. $A = (1,0,1)$ $\quad \vec{AB} = B - A = (0, 1, -1)$

$B = (1,1,0)$ $\quad \vec{AC} = C - A \ (0, -3, 3)$

$C = (1,-3,4)$

$\vec{AB} = K\,\vec{AC}$ condição para ser colinear

$(0, 1, -1) = K(0, -3, 3) \Leftrightarrow$

$\Leftrightarrow K = -\frac{1}{3}$ //

Para descobrir a equação vetorial da reta usamos um ponto e um vetor ambos da reta

$(x, y, z) = (1, 0, 1) + K(0, 1, -1)$

13. $\vec{CD} = D - C = (2, -6, 4)$

$(x, y, z) = (2, 1, -1) + K(2, -6, 4) \Leftrightarrow$

$\Leftrightarrow (x, y, z) = (2+2K, 1-6K, -1+4K)$ //

$\Leftrightarrow (3, -2, 1) = (2+2K, 1-6K, -1+4K) \Leftrightarrow$

$\Leftrightarrow \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right) = (K, K, K)$ $\quad A \in \vec{CD}$ //

$(-1, 2, 0) = (2+2K, 1-6K, -1+4K) \Leftrightarrow$

$\Leftrightarrow \left(-\frac{3}{2}, -\frac{1}{6}, \frac{1}{4}\right) = (K, K, K)$

14. $(x, y, z) = (3, 1, -1) + K(1, -1, 2) \Leftrightarrow$

$$\Leftrightarrow \begin{cases} x = 3 + K \\ y = 1 - K \\ z = -1 + 2K \end{cases}$$

15. $x + y + z = d \Leftrightarrow$ $\quad \boxed{x + y + z - 3 = 0}$

$\Leftrightarrow d = 3$

16. $A = (1, 1, 0)$ $\quad \vec{AB} = (5, -1, -1)$

$B = (6, 0, -1)$ $\quad \vec{AC} = (2, -1, 0)$

$C = (3, 0, 0)$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 5 & -1 & -1 \\ 2 & -1 & 0 \end{vmatrix} = (-1, 2, -3)$$

$-x + 2y - 3z + d = 0$

Substituindo o ponto A:

$d = -1$

$$\boxed{-x + 2y - 3z - 1 = 0}$$

17 $x(x-2) - 4(7-z) = -(y^2 + z^2) \Leftrightarrow$

$\Leftrightarrow x^2 - 2x - 28 + 4z = -y^2 - z^2 \Leftrightarrow$

$\Leftrightarrow (x-1)^2 + (y-0)^2 + (z+2)^2 = 33$

$C(1, 0, -2)$

$R = \sqrt{33}$